



Digital Design Verification

Exercise # 03

RISC-V – Procedures

Implementing Recursive Procedures

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Revision History

Revision Number	Revision Date	Revision By	Nature of Revision	Approved By
1.0	30/05/2024	Muhammad Imran	Complete manual	-



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Objective

The objective of this lab is to:

- Understand and analyze the implementation of a recursive procedure in RISC-V assembly language.

Tools

- Venus

Task 1

Implement the following recursive C language procedure in RISC-V assembly language.

```
int fact (int n)
{
    if (n < 1) return (1);
    else return (n * fact(n - 1));
}
```

- x10 = n (input argument)
- Returned value is also in x10
- May use x5 as temporary register!
- Multiply instruction:
 - **mul x10, x10, x6**
 - **x10 = x10 x x6**



Task 2

Implement the following recursive C language procedure in RISC-V assembly language.

```
int fib(int n){  
    if (n==0)  
        return 0;  
    else if (n == 1)  
        return 1;  
    else  
        return fib(n-1) + fib(n-2);  
}
```

- x10 = n (input argument)
- Returned value is also in x10